INCH-POUND

A-A-55540
11 DEC 1998
SUPERSEDING
MIL-R-13877D
03 APR 1987

COMMERCIAL ITEM DESCRIPTION

REGULATOR, OXYGEN, ACETYLENE, WELDING APPLICATION

The General Services Administration has authorized the use of this commercial item description (CID) as a replacement for Military Specification MIL-R-13877D for all federal agencies.

1. SCOPE. This commercial item description covers acetylene and oxygen gas regulators of the standard commercial types, which are attached to compressed gas cylinders, and used with oxy-acetylene welding and cutting torches, metallizing torches, laboratory equipment, etc. The regulators are for use with all types of torches. The single stage regulators are used where an occasional adjustment of the delivery pressure is required to maintain a constant delivery pressure, this adjustment not being detrimental to the operation. The two-stage regulators are used when an unattended constant delivery pressure is required with decreasing inlet pressures.

2. CLASSIFICATION. The gas regulators covered by this CID shall be of the following types:

Type Number	Description
I	Acetylene, single stage
II	Oxygen, single stage
III	Oxygen, two stage

Beneficial comments, recommendations, additions, deletions, clarifications, etc., and any data which may improve this document should be sent to: Defense Supply Center Columbus, ATTN: DSCC-VAI, Columbus, Ohio 43216-5000

FSC 4820

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:51Z
Check the source to verify that this is the current version before use.

3. SALIENT CHARACTERISTICS.

3.1 General. All acetylene and oxygen gas regulators shall meet the requirements of this CID. The regulators shall be standard commercial, manually adjusted, spring loaded diaphragm regulators. All gas regulators also shall:

a. Meet all requirements of Compressed Gas Association standard CGA E-4 titled Standard for Gas Pressure Regulators.

b. Meet all requirements of Underwriters Laboratories standard UL 252 titled Compressed Gas Regulators.

c. Meet all requirements of Underwriters Laboratories standard UL 252A titled Compressed Gas Regulator Accessories.

d. Have no materials which result in galvanic corrosion action, or shall be suitably protected against galvanic corrosion.

e. Have no self-initiated audible chatter or vibration over full operational conditions.

f. Have a vented bonnet.

g. Have a pressure responsive element which is replaceable without the application of heat.

h. Have pressure adjusting screws which are hand operable.

i. Have inlet and delivery pressure gauges which are in accordance with American National Standards Institute standard ANSI B40.1 titled Gauges - Pressure, Indicating Dial Type - Elastic Element, which are 2 inches in diameter or larger, and which are graduated in English and metric units.

j. Be suitable for outdoor field use.

k. Be resistant to weather, dust, and 99% humidity.

l. Have a replaceable, corrosion resistant filter which is compatible with the gas and pressure of intended service, which is of 60 micron pore size or less, and which is located in the inlet passage before the valve seat.

3.2 Characteristics. The regulators shall be new and of the manufacturer's current configuration, and shall have the characteristics included below.

3.2.1 Performance. The gas regulators shall have the performance characteristics shown in Table I. The flow characteristics requirements shall be read from this table as per the following example for the acetylene flow rate:

> "Acetylene regulators shall deliver acetylene at no less than 10 pounds per square inch gauge (psig) delivery pressure when the inlet pressure is no higher than 50 psig and the flow rate is no less than 275 standard cubic feet per hour (scfh)."

Flow capacities shall be determined as specified in CGA E-4. Standard cubic feet per hour (scfh) shall be the flow in cubic feet per hour at one atmosphere and 70 degrees Fahrenheit.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:51Z
Check the source to verify that this is the current version before use.

Table I. Regulator performance characteristics

Regulator	Maximum Inlet pressure, psig	Delivery pressure range, psig	Flow rate, scfh	Test pressure	
				Inlet, psig max	Outlet, psig min
Type I acetylene single stage	400	0 - 15, redlined to 30	275	50	10
Type II oxygen single stage	3,000	0 - 120 or higher	1,200	300	120
Type III oxygen double stage	3,000	0 - 120 or higher	1,200	300	120

3.2.2 Pressure regulation. The gas regulators shall have the pressure regulation performance shown in Table II.

Table II. Regulator pressure regulation

Regulator	Set inlet pressure to min of (psig)	Set flow to (scfh)	Change inlet pressure to max of (psig)	Max allowed outlet pressure change (psig)
Type I acetylene single stage	350	30 ± 15	50	2.4
Type II oxygen single stage	2,500	75 ± 25	500	16.0
Type III oxygen double stage	2,500	75 ± 25	500	2.0

3.2.3 Flow regulation. The gas regulators shall have the flow regulation performance shown in Table III.

Table III. Regulator flow regulation

Regulator	Set inlet pressure to max of (psig)	Set delivery pressure to (psig)	Set flow to (scfh)	Increase flow to minimum (scfh)	Maximum allowed delivery pressure change (psig)
Type I acetylene single stage	50 (cylinder may be manifold)	14 - 15	30 ± 15	275	5
Type II oxygen single stage	300	125 + 5	75 ± 25	1,200	15
Type III oxygen double stage	300	125 ± 5	75 ± 25	1,200	15

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:51Z
Check the source to verify that this is the current version before use.

3.2.4 Static increment. The gas regulators shall have the static increment performance shown in Table IV.

Table IV. Regulator static increment performance

Regulator	Set inlet pressure to minimum of (psig)	Set flow to (scfh)	Change flow to (scfh)	Maximum delivery pressure increase (psig)	
				First 2 minutes	Next 5 minutes
Type I acetylene single stage	50	30 ± 15	0	5	0
Type II oxygen single stage	300	75 ± 25	0	10	0
Type III oxygen double stage	300	75 ± 25	0	10	0

3.3 Acetylene regulator maximum delivery pressure. The acetylene regulators shall be designed to allow the maximum delivery pressure to be no greater than 30 psig.

3.4 Inlet connections. Inlet connections shall be as specified below, unless otherwise specified in the contract.

3.4.1 Acetylene inlet connection. Acetylene regulators shall have an inlet connection which complies with connection number 510 of Compressed Gas Association standard CGA V-1 titled Standard for Compressed Gas Cylinder Valve Outlet and Inlet Connections. Acetylene regulators shall also be available with a CGA V-1 number 300 inlet connector if specified in contract. The use of an adapter is acceptable for this connection.

3.4.2 Oxygen inlet connection. Oxygen regulators shall have an inlet connection which complies with connection number 540 of Compressed Gas Association standard CGA V-1 titled Standard for Compressed Gas Cylinder Valve Outlet and Inlet Connections.

3.5 Outlet connections. Outlet connections shall be as specified below, unless otherwise specified in the contract.

3.5.1 Acetylene outlet connection. Acetylene regulators shall have an outlet connection which complies with connection number 023 of Compressed Gas Association standard CGA E-1 titled Standard Connections for Regulator Outlets, Torches, and fitted Hose for Welding and Cutting Equipment.

3.5.2 Oxygen outlet connection. Oxygen regulators shall have an outlet connection which complies with connection number 022 of Compressed Gas Association standard CGA E-1 titled Standard Connections for Regulator Outlets, Torches, and fitted Hose for Welding and Cutting Equipment.

3.6 Relief valve.

3.6.1 Acetylene regulator relief valve. Relief valves are not required for acetylene regulators.

3.6.2 Single stage oxygen regulator. Single stage oxygen regulators shall have at least one self-resetting relief valve in the regulator. The relief valve shall activate at 200 ± 50 psig from the delivery side of the regulator.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:51Z
Check the source to verify that this is the current version before use.

3.6.3 Two stage oxygen regulator. Two stage oxygen regulators shall have at least one self-resetting relief valve in the regulator. The relief valve shall activate at 200 ± 50 psig from the delivery side of the regulator. However, if a relief valve is installed in the first stage of a two stage regulator, the valve shall be set to relieve pressure according to the manufacturer's standard practice to protect the regulator from over pressure and no valve is then required to be installed in the second stage.

3.7 Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within specified tolerances using conversion tables contained in the latest revision of FED-STD-376, entitled Preferred Metric Units for General Use by the Federal Government, and all other requirements of this commercial item description are met.

3.8 Marking. Products supplied to this CID shall be marked with the manufacturer's standard commercial part number(s) and with Underwriters Laboratory marking.

4. REGULATORY REQUIREMENTS.

4.1 Recovered Materials. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

4.2 Environmental Protection. During the manufacture of the gas regulators, materials hazardous to the ecological systems as prohibited by the Federal, State or local statutes in effect on the date of the contract shall not be used or emitted.

5. QUALITY ASSURANCE.

5.1 Product Conformance. The products provided shall meet the salient characteristics of this CID, conform to the producer's own drawings, specifications, standards and quality assurance practices, and be the same product offered for sale in the commercial market. The Government reserves the right to require proof of such conformance.

5.2 Market Acceptance. The following market acceptance criteria are necessary to assure the quality of the products to be provided under the requirements of this CID.

a. The company must have produced the product meeting the requirements of this CID for at least five years.

b. The company must have sold at least an average of 1,000 units per year of the product meeting the requirements of this CID in the commercial market placeand/or to the government over the last five years, will no more than 50% of these sales going to the government.

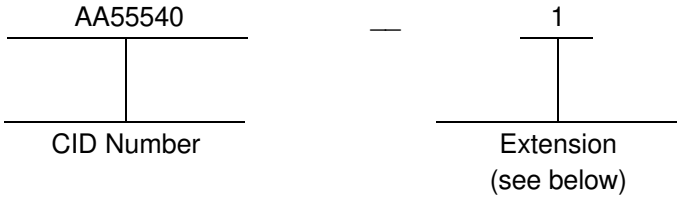
6. PACKAGING.

6.1 Preservation, packing and marking. Preservation, packing, and marking shall be as specified in the contract or order.

7. NOTES

7.1 Part identification number (PIN). The following part identification numbering procedure is for government purposes and does not constitute a requirement for the contractor. The gas regulators described by this CID use a PIN as shown in the following example:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:51Z
Check the source to verify that this is the current version before use.



Extension	Description
1	Acetylene, single stage (Type I)
2	Oxygen, single stage (Type II)
3	Oxygen, two stage (Type III)

7.2 PIN-National Stock Number (NSN) Cross Reference. The cross reference matrix between PIN and NSN for the gas regulators is shown in Table V.

Table V. PIN – NSN Cross reference matrix

Type	Description	PIN	NSN
I	Acetylene-Single Stage	AA55540-1	4820-00-285-6067 4820-00-551-1094
II	Oxygen-Single Stage	AA55540-2	4820-01-256-0264
III	Oxygen-Double Stage	AA55540-3	4820-00-641-3519

7.3 Ordering data. The contract or order should specify the following:

a. CID document number, revision, and CID PIN
b. Quality assurance provisions
c. Packaging requirements
d. Non-standard equipment characteristics, such as special adapters for inlet or outlet connections

7.4 Commercial products. At the time of CID preparation and coordination, the vendors listed in Table VI were known to manufacture or sell commercial products which nominally meet the requirements of this CID. (NOTE: This information should not be considered as a list of approved manufacturers or be used to restrict procurement to only the manufacturers shown.)

Table VI. Gas regulator manufacturers

CAGE Code	Name and address
0A389	Controls Corporation of America(Concoa) 1501 Harpers Road Virginia Beach, VA 23454 Tel: 1-757-422-8330 Fax: 1-757-422-3125

Table VI. Gas regulator manufacturers - continued

CAGE Code	Name and address
96073	Esab, Inc. 411 S. Ebenezer Road Florence, SC 29501-0545

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:51Z
Check the source to verify that this is the current version before use.

	Tel: 1-803-669-4411 Fax: 1-803-664-4258
55681	Smith Equipment 2601 Lockheed Avenue Watertown, SD 57201 Tel: 1-800-328-3363 Fax: 1-800-685-3370
17941	Uniweld Products Inc. 2850 Ravenswood Road Ft. Lauderdale, FL 33312-4994 Tel: 1-954-584-2000 Fax: 1-954-587-0109
63026	Victor Equipment Company P.O. Box 1007 Denton, TX 76202-1007 Tel: 1-800-426-1888 Fax: 1-800-535-0557
0A1K0	Western Enterprises 875 Bassett Road Westlake, OH 44145-1196 Tel: 1-216-871-2160 Fax: 1-216-835-8283

7.5 Referenced documents.

CGA E-4 titled Standard for Gas Pressure Regulators is available from the Compressed Gas Association, 1725 Jefferson Davis Highway, Arlington, VA 22202-4102

CGA V-1 titled Standard for Compressed Gas Cylinder Valve Outlet and Inlet Connections is available from the Compressed Gas Association, 1725 Jefferson Davis Highway, Arlington, VA 22202-4102

CGA E-1 titled Standard Connections for Regulator Outlets, Torches, and fitted Hose for Welding and Cutting Equipment is available from the Compressed Gas Association, 1725 Jefferson Davis Highway, Arlington, VA 22202-4102

UL 252 titled Compressed Gas Regulators is available from Underwriters Laboratories, 1285 Walt Whitman Road, Melville, New York 11747-3081

UL 252A titled Compressed Gas Regulator Accessories is available from Underwriters Laboratories, 1285 Walt Whitman Road, Melville, New York 11747-3081

ANSI B40.1 titled Gauges - Pressure, Indicating Dial Type - Elastic Element is available from American National Standards Institute, 1430 Broadway, New York, NY 10018

FED-STD-376, titled Preferred Metric Units for General Use by the Federal Government, is available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:51Z
Check the source to verify that this is the current version before use.

Federal Acquisition Regulations (FAR) may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

Unless otherwise specified in the contract, the issues of the documents referenced in this CID shall be the issues which are current as of the date of the solicitation for equipment purchased under this CID.

MILITARY INTERESTS:

Custodians:
Army – AL
Air Force – 99

Review activities:
Air Force – 82

Preparing activity:
DLA – CC

(Project No. 4820-0759-01)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:51Z
Check the source to verify that this is the current version before use.